BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 046 · Lesson 046 · Objects in Arrays and Lists

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a constructor used for?

Q2. What does private mean for an instance field?

Q3. What index identifies the first C# array item?

Q4. 031

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

object — Instance of a class.

reference — Value identifying an object instance.

aliasing — Several references to the same object.

index — Position in a sequence.

New learning

Observe — A class describes an object; the array stores references to two Book instances.

Explain — Indexes start at 0. shelf[1] identifies the second reference, then .Title reads its property.

Apply — An array has a fixed length after creation; individual array elements can refer to objects whose properties change.

Worked example

Trace the title stored at books[0] in a two-object Book array.

1. Trace and justify — Index 0 refers to the first object; its Title is Ada.

We Do · Practise with guidance

In a different array, which title prints after updating the object at index 1?

```
Book[] b={new Book{Title="One"},new Book{Title="Two"}};
b[1].Title="Updated";
Console.WriteLine(b[1].Title);
```

Trace and justify

Index 1 refers to the second object. The property update changes its Title; output is Updated.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Given shelf titles Dune,Holes, give output of shelf[1].Title and reason. [4]

Question 2

Write C# to create an array of two Book objects with titles A and B, then print first title. [6]

Question 3

Two array elements reference the SAME Book object. Changing its title via first reference is then printed via second. Explain. [6]

Exit Ticket · Answer before checking

Updating a property through one reference can be visible through another alias.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Initialising an object's state.

Q2. It cannot be accessed directly from outside the class.

Q3. Zero.

Q4. A rising-edge D flip-flop has Q=0 and D=1 just before the next active edge. Give Q after it and why.

You Do pages · Independent questions

Q1. Holes [2]; 0-based index 1 selects second element and its Title [2].

Q2. Book[] b={new Book{Title="A"},new Book{Title="B"}}; [4] Console.WriteLine(b[0].Title); [2], assuming Book declares Title property.

Q3. Both elements hold references [2] to same instance [2]; mutation visible through either reference [2]; this is aliasing, not array-index copying.

Exit Ticket · Model answer

True. Both can point to the same instance.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: